RHEL 9.7 Nginx Local Repository Setup

Command Reference & Implementation Log

Server: 192.168.121.160 | Date: April 11, 2026

1. Mount ISO & Copy Repository Files

Prepare the RHEL 9.7 DVD ISO as the repository source by mounting it and copying contents.

1	`mkdir -p /var/www/html/rhel97`	Create web root directory for repository
2	`mount -o loop rhel-9.7-x86_64-dvd.iso /mnt`	Mount ISO image as loop device
3	`cp -r /mnt/* /var/www/html/rhel97/`	Copy all ISO contents to web directory
4	`umount /mnt`	Unmount the ISO after copying

2. Install & Enable Nginx

Install the Nginx web server and configure it to start automatically.

5	`dnf install nginx -y`	Install Nginx via DNF package manager
6	`systemctl enable --now nginx`	Enable and start Nginx service immediately
7	`systemctl status nginx`	Verify Nginx is active and running

3. Generate SSL Certificate

Create a self-signed SSL certificate for HTTPS support.

8	`mkdir -p /etc/nginx/ssl`	Create SSL directory for certificates

9	`openssl req -new -x509 -days 3650 -nodes \  -out /etc/nginx/ssl/server.crt \  -keyout /etc/nginx/ssl/server.key`	Generate self-signed certificate valid for 10 years

4. Configure Nginx Virtual Host

Create the Nginx configuration file to serve the RHEL 9.7 repository over both HTTP and HTTPS.

10	`vi /etc/nginx/conf.d/RHEL-9.7-nginx-repo.conf`	Create new Nginx virtual host config file

Final working configuration content:

```
# HTTP block for the installer
server {
    listen 80;
    server_name 192.168.121.160;

    location = /rhel97 {
        return 301 http://$host/rhel97/;
    }

    location /rhel97/ {
        autoindex on;
        alias /var/www/html/rhel97/;
    }
}

# HTTPS block
server {
    listen 443 ssl;
    server_name 192.168.121.160;
    ssl_certificate /etc/nginx/ssl/server.crt;
    ssl_certificate_key /etc/nginx/ssl/server.key;

    location = /rhel97 {
        return 301 https://$host/rhel97/;
    }

    location /rhel97/ {
```

```
        autoindex on;
        alias /var/www/html/rhel97/;
    }
}
```

11	`nginx -t`	Test Nginx configuration for syntax errors
12	`mv rhel9.conf rhel9.conf-2026`	Disable old conflicting config (rename to non-.conf)
13	`systemctl reload nginx`	Reload Nginx to apply new configuration

5. Fix SELinux Permissions

The repository files had the wrong SELinux context (var_t instead of httpd_sys_content_t), causing 403 Forbidden errors. Fix by relabeling.

14	`ls -lZ /var/www/html/rhel97/BaseOS/repodata/repomd.xml`	Check current SELinux context on files
15	`chcon -Rt httpd_sys_content_t /var/www/html/rhel97/`	Recursively set correct SELinux context

Verification — both commands should return HTTP 200 OK:

16	`curl -I http://192.168.121.160/rhel97/BaseOS/repodata/repomd.xml`	Verify BaseOS repo is accessible (expect 200 OK)
17	`curl -I http://192.168.121.160/rhel97/AppStream/repodata/repomd.xml`	Verify AppStream repo is accessible (expect 200 OK)

6. Configure Firewall

Open HTTP (port 80) in addition to the already-open HTTPS (port 443) so DNF clients can reach the repository.

18	`firewall-cmd --permanent --add-service=http`	Permanently allow HTTP traffic (port 80) through firewall
19	`firewall-cmd --permanent --add-service=https`	Permanently allow HTTPS traffic (port 443) through firewall
20	`firewall-cmd --reload`	Reload firewall to apply changes
21	`firewall-cmd --list-all`	Confirm both http and https are listed in services

Verify HTTPS repository access (self-signed cert — use -k to skip cert validation):

22	`curl -kI https://192.168.121.160/rhel97/BaseOS/repodata/repomd.xml`	Verify BaseOS repo accessible over HTTPS (expect 200 OK)
23	`curl -kI https://192.168.121.160/rhel97/AppStream/repodata/repomd.xml`	Verify AppStream repo accessible over HTTPS (expect 200 OK)

7. Final Verification

24	`nginx -t && systemctl reload nginx`	Final config test and reload
25	`systemctl status nginx`	Confirm Nginx is active (running)

Notes & Troubleshooting

Key Issues Encountered & Resolved

- 404 Errors: Initially only HTTPS (port 443) was configured. Added HTTP (port 80) server block to RHEL-9.7-nginx-repo.conf.
- 403 Forbidden: Files had SELinux context var_t. Fixed with chcon -Rt httpd_sys_content_t /var/www/html/rhel97/.
- Conflicting server_name warning: Old rhel9.conf also used port 443 for same IP. Resolved by renaming it to rhel9.conf-2026 so Nginx ignores it.

- Firewall: HTTP service was missing. Added with firewall-cmd --permanent --add-service=http.

Repository URL for DNF Clients

Use the following base URLs in /etc/yum.repos.d/*.repo on client machines:

HTTP (no SSL validation needed):

```
baseurl=http://192.168.121.160/rhel97/BaseOS/
baseurl=http://192.168.121.160/rhel97/AppStream/
```

HTTPS (add sslverify=0 for self-signed cert):

```
baseurl=https://192.168.121.160/rhel97/BaseOS/
baseurl=https://192.168.121.160/rhel97/AppStream/
sslverify=0
```